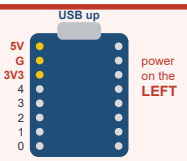
Top view, enlarged for reading

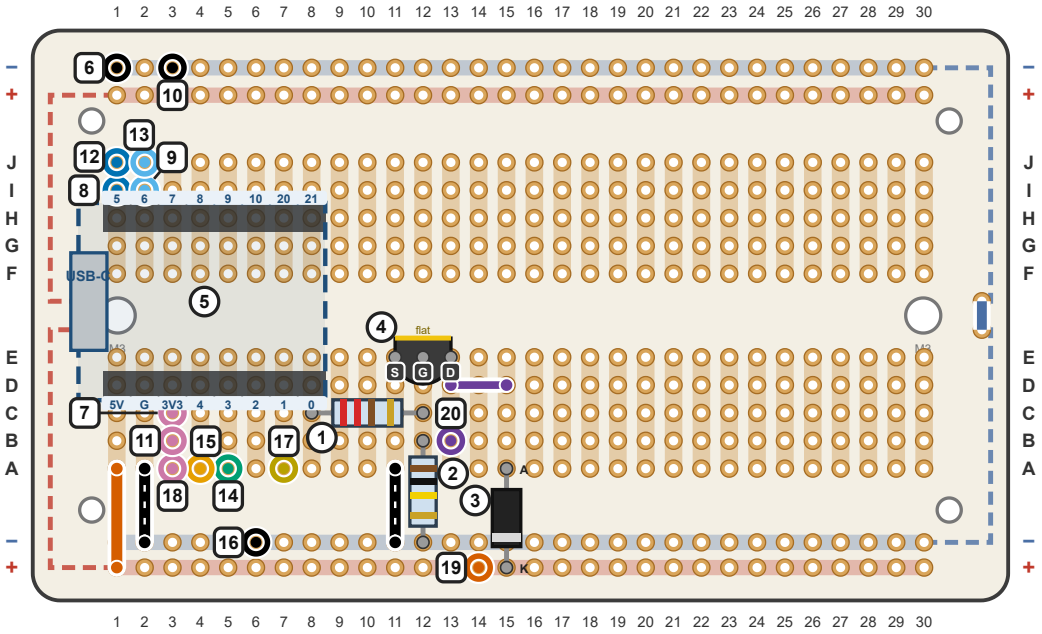
Printer station, C3 POWER-LEFT — protoboard

TOP VIEW (component side), enlarged, not to scale

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C8, C12
2. RP1 100 kΩ: B12, bottom − rail at number 12
3. D1 1N4001: bottom + rail at number 15, A15
4. Q1 BS170: E13, E12, E11
5. U1 ESP32-C3 Super Mini: D1-D8, H1-H8

Device wires

PN532

6. GND: top − rail at number 1
7. 3V3: C3
8. SDA: I1
9. SCL: I2

OLED

10. GND: top − rail at number 3
11. 3V3: B3
12. SDA: J1
13. SCL: J2

encoder

14. CLK: A5
15. DT: A4
16. GND: bottom − rail at number 6
17. SW: A7
18. 3V3: A3

buzzer

19. 5V: bottom + rail at number 14
20. BZ−: B13

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

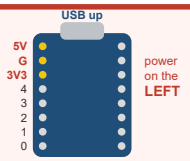
Solder side, enlarged for reading

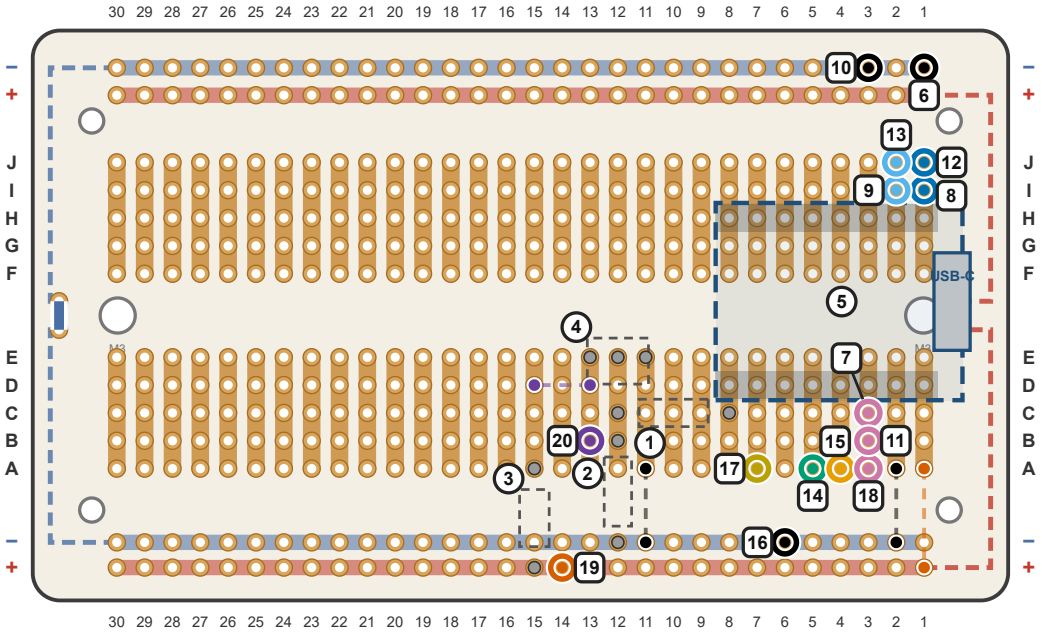
Printer station, C3 POWER-LEFT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C8, C12
2. RP1 100 kΩ: B12, bottom − rail at number 12
3. D1 1N4001: bottom + rail at number 15, A15
4. Q1 BS170: E13, E12, E11
5. U1 ESP32-C3 Super Mini: D1-D8, H1-H8

Device wires

PN532

6. GND: top − rail at number 1
7. 3V3: C3
8. SDA: I1
9. SCL: I2

OLED

10. GND: top − rail at number 3
11. 3V3: B3
12. SDA: J1
13. SCL: J2

encoder

14. CLK: A5
15. DT: A4
16. GND: bottom − rail at number 6
17. SW: A7
18. 3V3: A3

buzzer

19. 5V: bottom + rail at number 14
20. BZ−: B13

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- [9] a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

This bar is NOT 20 mm on this enlarged page: measure on the real-size pages

Real size (print at 100 %, "actual size"): lay the board on the drawing

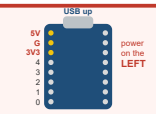
Check the 20 mm bar with a ruler before you rely on the drawing.

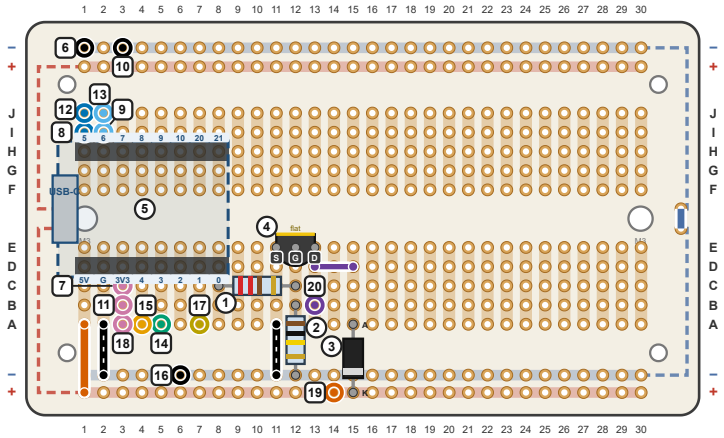
Printer station, C3 POWER-LEFT — protoboard

TOP VIEW (component side), real size

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C8, C12
2. RP1 100 kΩ: B12, bottom − rail at number 12
3. D1 1N4001: bottom + rail at number 15, A15
4. Q1 BS170: E13, E12, E11
5. U1 ESP32-C3 Super Mini: D1-D8, H1-H8

Device wires

PN532

6. GND: top − rail at number 1
7. 3V3: C3
8. SDA: I1
9. SCL: I2

OLED

10. GND: top − rail at number 3
11. 3V3: B3
12. SDA: J1
13. SCL: J2

encoder

14. CLK: A5
15. DT: A4
16. GND: bottom − rail at number 6
17. SW: A7
18. 3V3: A3

buzzer

19. 5V: bottom + rail at number 14
20. BZ−: B13

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

- wire end: solder here
- copper strip: its holes are joined
- (4) a part on the board (see Parts)
- (9) a wire of an off-board device (see Device wires)
- unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

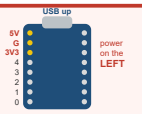
Real size, solder side

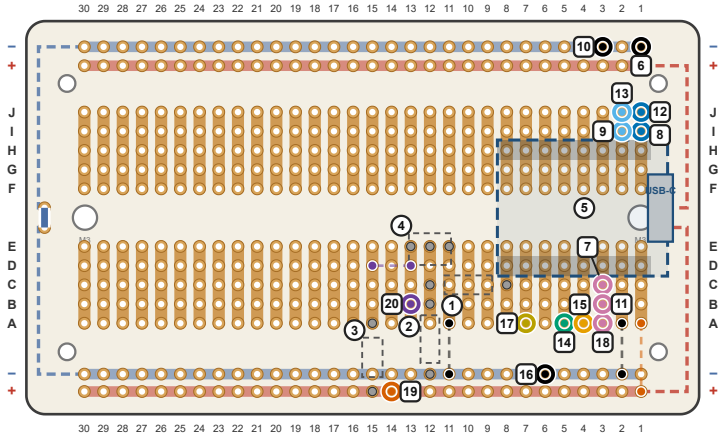
Printer station, C3 POWER-LEFT — protoboard

SOLDER SIDE (mirrored): number 1 is on the RIGHT

For a C3 POWER-LEFT module only

Hold the module USB up, chip facing you: 5V, G and 3V3 must be the top three pins of the LEFT row. A power-right module here puts 5 V on a GPIO pin and can destroy it. Check before you plug U1 in.





Parts

1. RG1 220 Ω: C8, C12
2. RP1 100 kΩ: B12, bottom − rail at number 12
3. D1 1N4001: bottom + rail at number 15, A15
4. Q1 BS170: E13, E12, E11
5. U1 ESP32-C3 Super Mini: D1-D8, H1-H8

Device wires

PN532

6. GND: top − rail at number 1
7. 3V3: C3
8. SDA: I1
9. SCL: I2

OLED

10. GND: top − rail at number 3
11. 3V3: B3
12. SDA: J1
13. SCL: J2

encoder

14. CLK: A5
15. DT: A4
16. GND: bottom − rail at number 6
17. SW: A7
18. 3V3: A3

buzzer

19. 5V: bottom + rail at number 14
20. BZ−: B13

Nets (wire colours)

- **GND** ground
- **3V3** 3.3 V from the C3 (power)
- **SCL** I2C clock, GPIO 6
- **DT** encoder B, GPIO 4
- **BUZ** buzzer drive, GPIO 0
- **BZ−** buzzer − / Q1 drain
- **5V** USB 5 V (power)
- **SDA** I2C data, GPIO 5
- **CLK** encoder A, GPIO 3
- **SW** encoder switch, GPIO 1
- **GATE** Q1 gate

wire end: solder here

copper strip: its holes are joined

(4) a part on the board (see Parts)

(9) a wire of an off-board device (see Device wires)

unmarked edge trace, broken at the link pads beside the M3 holes; the links join the halves

20 mm: printed at 100 % (actual size) this bar measures 20 mm and the board is real size

Build order and hole table

For a C3 POWER-LEFT module only (USB up: 5V, G, 3V3 at the top of the LEFT row). A power-right module in this layout puts 5 V on a GPIO pin and can destroy it.

Build order (lowest parts first). The numbers are the callouts on the drawing; wire links carry no number:

- Rail link (−): wire from the right link pad, upper to the right link pad, lower. It joins the top and bottom − rails.
- Wire 5V: A1 to bottom + rail at number 1 (straight, 9 mm).
- Wire GND: A2 to bottom − rail at number 2 (straight, 7 mm).
- Wire GND: A11 to bottom − rail at number 11 (straight, 7 mm).
- Wire BZ−: D13 to D15 (straight, 5 mm).
- Part 1, RG1 (220 Ω): C8 to C12, lying flat.
- Part 2, RP1 (100 kΩ): B12 to bottom − rail at number 12, lying flat.
- Part 3, D1 (1N4001): banded end (K) in bottom + rail at number 15, other end (A) in A15.
- Part 4, Q1 (BS170): D in E13, G in E12, S in E11, flat face toward the top edge (the J side).
- Part 5, U1 sockets: two 1x8 female headers in D1-D8, H1-H8. Plug U1 in while soldering to keep them square, then take it out.
- Device wires 6-9, PN532: 6 GND in top − rail at number 1, 7 3V3 in C3, 8 SDA in I1, 9 SCL in I2. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 10-13, OLED: 10 GND in top − rail at number 3, 11 3V3 in B3, 12 SDA in J1, 13 SCL in J2. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 14-18, encoder: 14 CLK in A5, 15 DT in A4, 16 GND in bottom − rail at number 6, 17 SW in A7, 18 3V3 in A3. Solder each wire from the top, then tie the bundle to the board edge.
- Device wires 19-20, buzzer: 19 5V in bottom + rail at number 14, 20 BZ− in B13. Solder each wire from the top, then tie the bundle to the board edge.
- Check with a meter: no short from 5V, 3V3 or SDA/SCL to GND. Then plug U1 in, USB-C toward the board edge as drawn, after checking it is a power-left module.

No.	Item	Holes
	− rail link	right link pad, upper to right link pad, lower
	wire 5V	A1 to bottom + rail at number 1
	wire GND	A2 to bottom − rail at number 2
	wire GND	A11 to bottom − rail at number 11
	wire BZ−	D13 to D15
1	RG1 220 Ω	C8 to C12
2	RP1 100 kΩ	B12 to bottom − rail at number 12
3	D1 1N4001	K bottom + rail at number 15, A A15
4	Q1 BS170	D E13, G E12, S E11
5	U1 sockets	D1-D8, H1-H8
6-9	PN532 wires	6 GND top − rail at number 1, 7 3V3 C3, 8 SDA I1, 9 SCL I2
10-13	OLED wires	10 GND top − rail at number 3, 11 3V3 B3, 12 SDA J1, 13 SCL J2
14-18	encoder wires	14 CLK A5, 15 DT A4, 16 GND bottom − rail at number 6, 17 SW A7, 18 3V3 A3
19-20	buzzer wires	19 5V bottom + rail at number 14, 20 BZ− B13